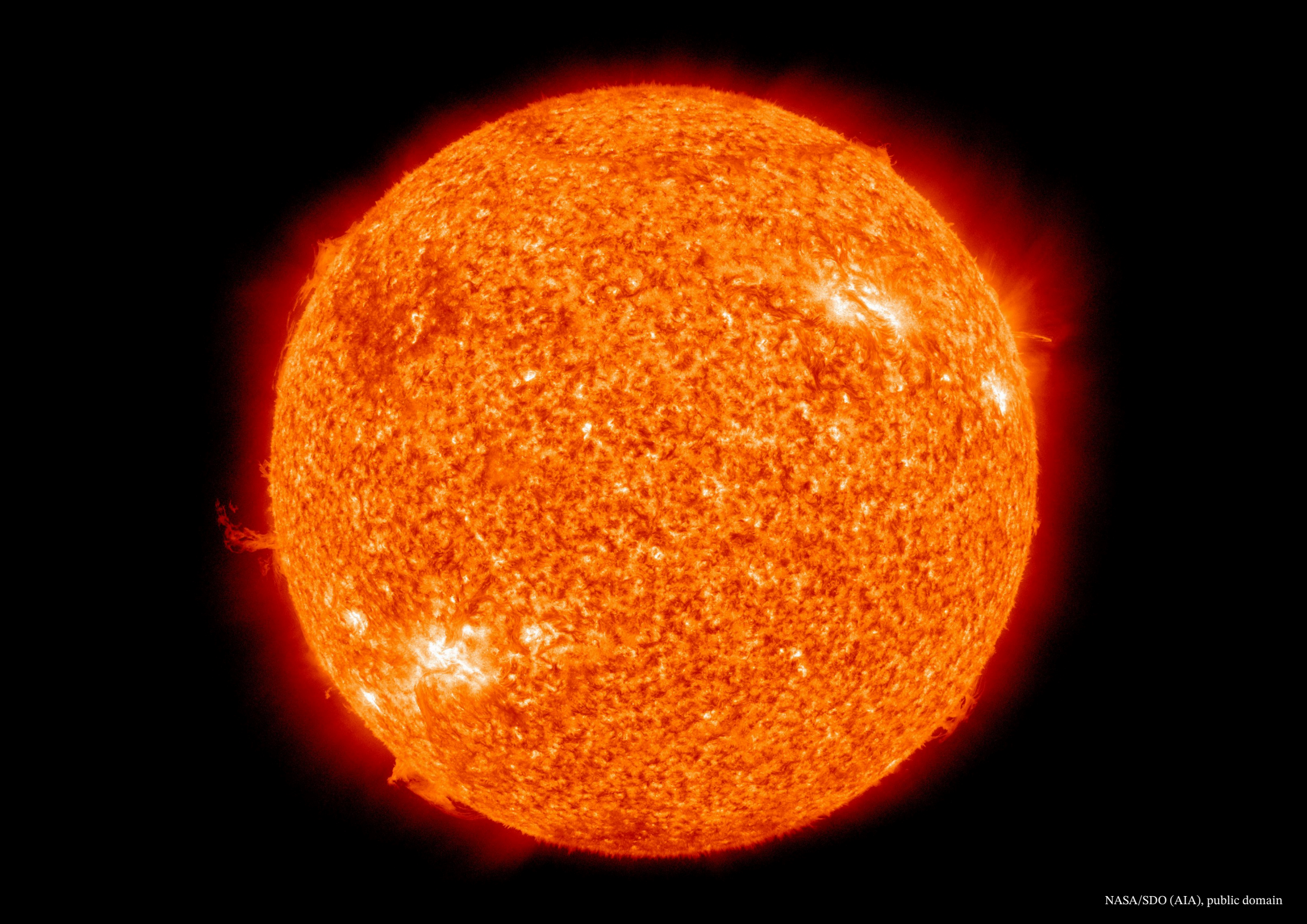
Fusion in stars

Christopher Albert



NASA/SDO (AIA), public domain

3.1 Energy budget of the Sun

Sun $L_\odot = 3.83 \times 10^{26}\ \mathrm{W}, \quad M_\odot = 1.99 \times 10^{30}\ \mathrm{kg}, \quad R_\odot = 6.96 \times 10^{8}\ \mathrm{m}$

gravity $t_{\mathrm{KH}} = \dfrac{GM_\odot^2}{R_\odot L_\odot} \approx 3 \times 10^{7}\ \mathrm{yr}$

fusion $4\,\mathrm{p} \to {}^4\mathrm{He}: \quad \Delta m/m = 0.71\%, \quad t_{\mathrm{nuc}} = \dfrac{0.1 \cdot 0.0071\, M_\odot c^2}{L_\odot} \approx 1 \times 10^{10}\ \mathrm{yr}$

Earth $4.5 \times 10^{9}\ \mathrm{yr}$ (radiometric)

$$t_{\mathrm{KH}} \ll 4.5\ \mathrm{Gyr} < t_{\mathrm{nuc}}$$

only nuclear energy lasts long enough

Eddington · Nature 106, 14 (1920) · Bethe · Phys. Rev. 55, 434 (1939)

3.2 Hydrostatic equilibrium and virial theorem

force balance $$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{Gm(r)\rho}{r^2}$$

virial theorem $$2K + U = 0, \quad K = \frac{3}{2}Nk_B\overline{T}, \quad U \approx -\frac{3}{5}\frac{GM^2}{R}$$

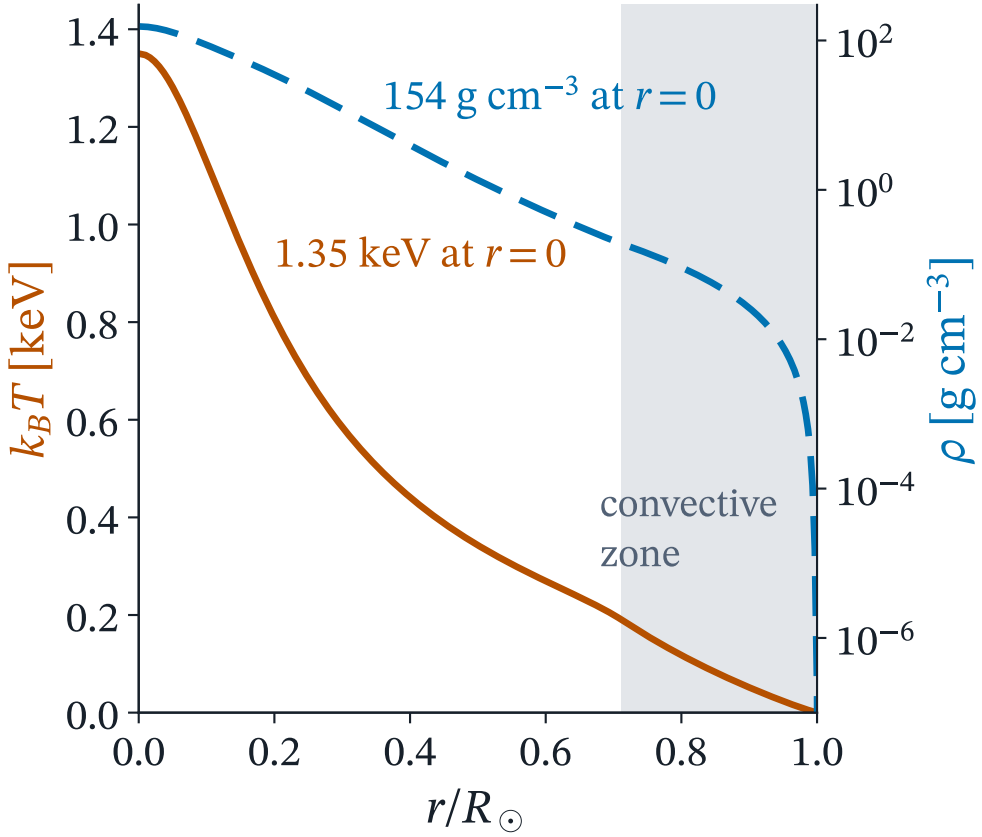
mean temperature $$k_B\overline{T} = \frac{GM\mu m_u}{5R} \approx 0.25\ \mathrm{keV}$$ ($\mu = 0.62$); centre 1.35 keV

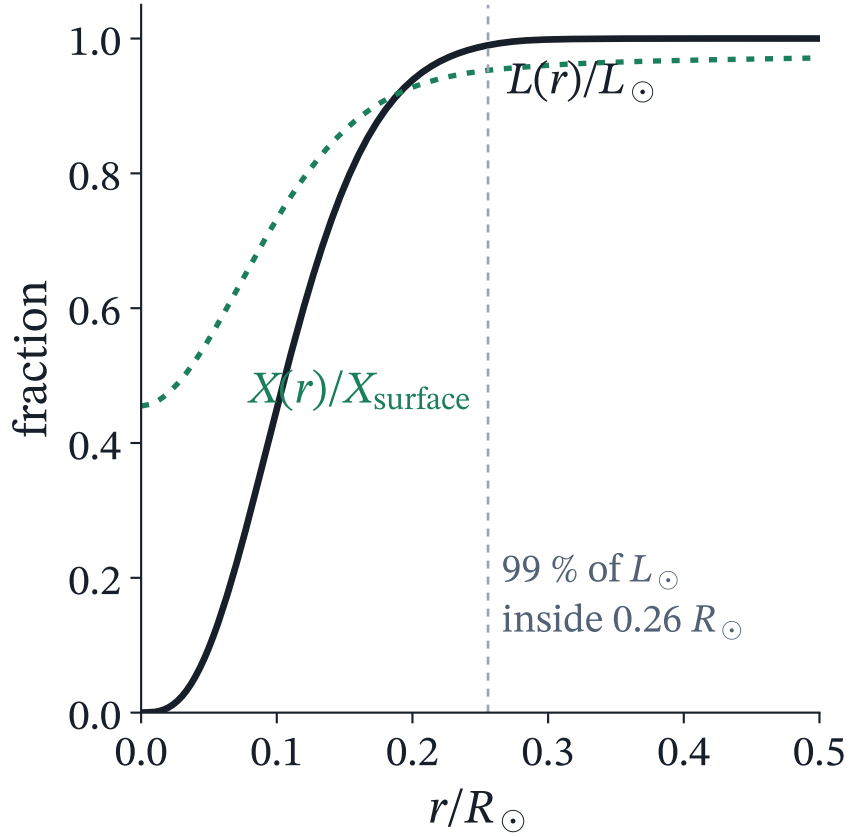
total energy $$E = K + U = -K \quad \Rightarrow \quad \frac{\mathrm{d}K}{\mathrm{d}E} = -1$$

losing energy heats the star

too much fusion: expands and cools · a thermostat

3.2 Inside the Sun





centre: $k_B T = 1.35$ keV, $\rho = 1.5 \times 10^5$ kg/m^3, $p_{\mathrm{fus}} = 262$ W/m^3

Model S · Christensen-Dalsgaard et al. · Science 272, 1286 (1996)

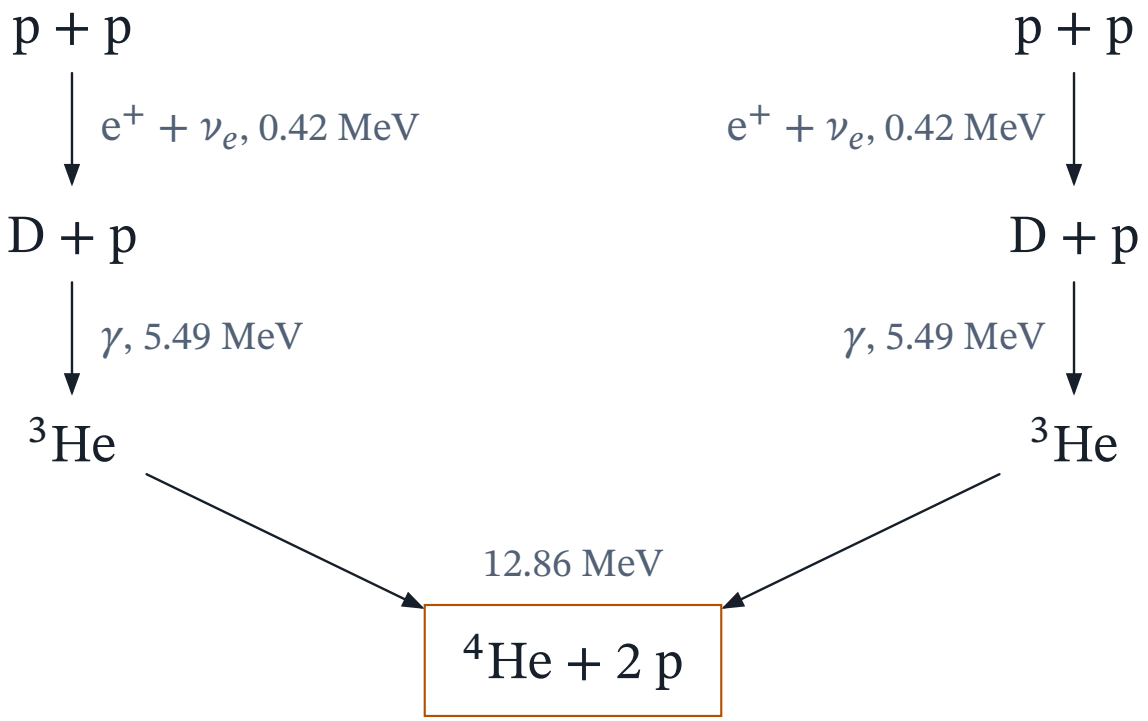
3.2 The Sun and a tokamak

	Sun, centre	tokamak (ITER)
$k_B T$	1.35 keV	10–20 keV
n	$4.6 \times 10^{31}\,\mathrm{m}^{-3}$	$10^{20}\,\mathrm{m}^{-3}$
fuel	p–p (weak interaction)	D–T
p_{fus}	$262\ \mathrm{W/m^3}$	$0.6\ \mathrm{MW/m^3}$
$\tau_E = W_{\mathrm{th}}/L$	2×10^7 yr	3.7 s
$nT\tau_E$	$5 \times 10^{46}\,\mathrm{keV\,s\,m^{-3}}$	needs $3 \times 10^{21}\,\mathrm{keV\,s\,m^{-3}}$

human body: about $1\ \mathrm{kW/m^3}$ Sun: fusion wins by τ_E, not by power density

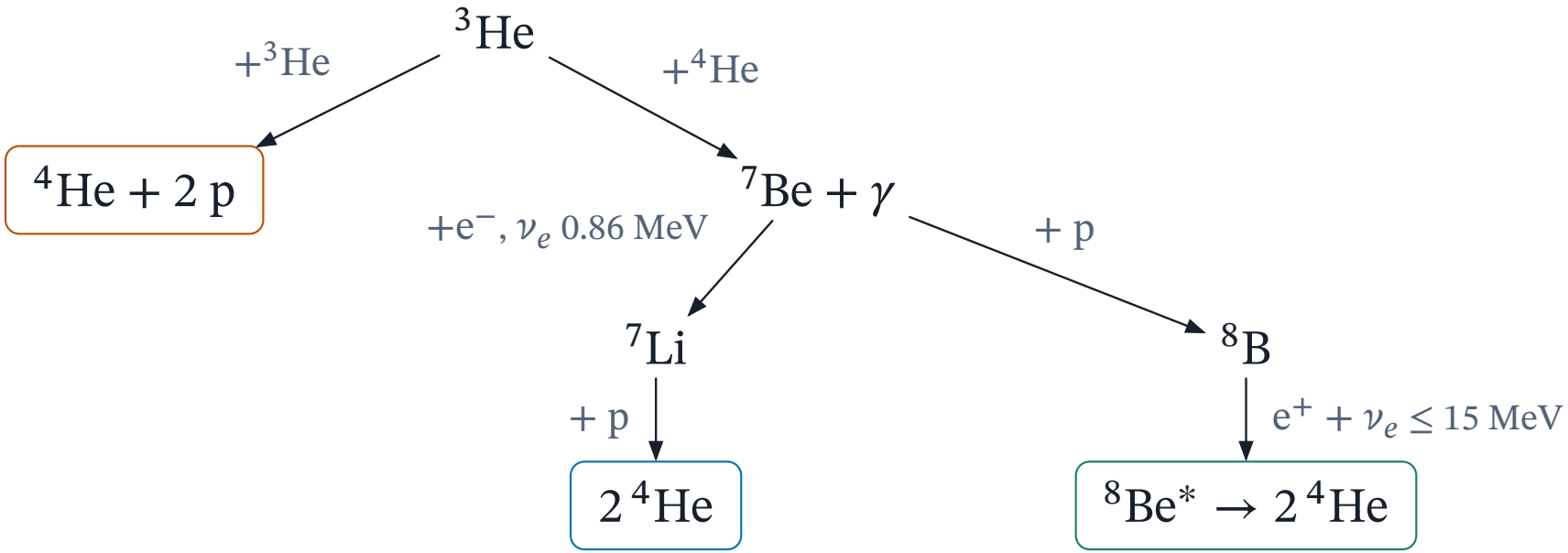
ITER Physics Basis · Nucl. Fusion 39, 2137 (1999)

3.3 Proton–proton chain



$4\,\mathrm{p} \rightarrow {}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e$ 26.7 MeV incl. $2 \times \mathrm{e}^+\mathrm{e}^-$ annihilation

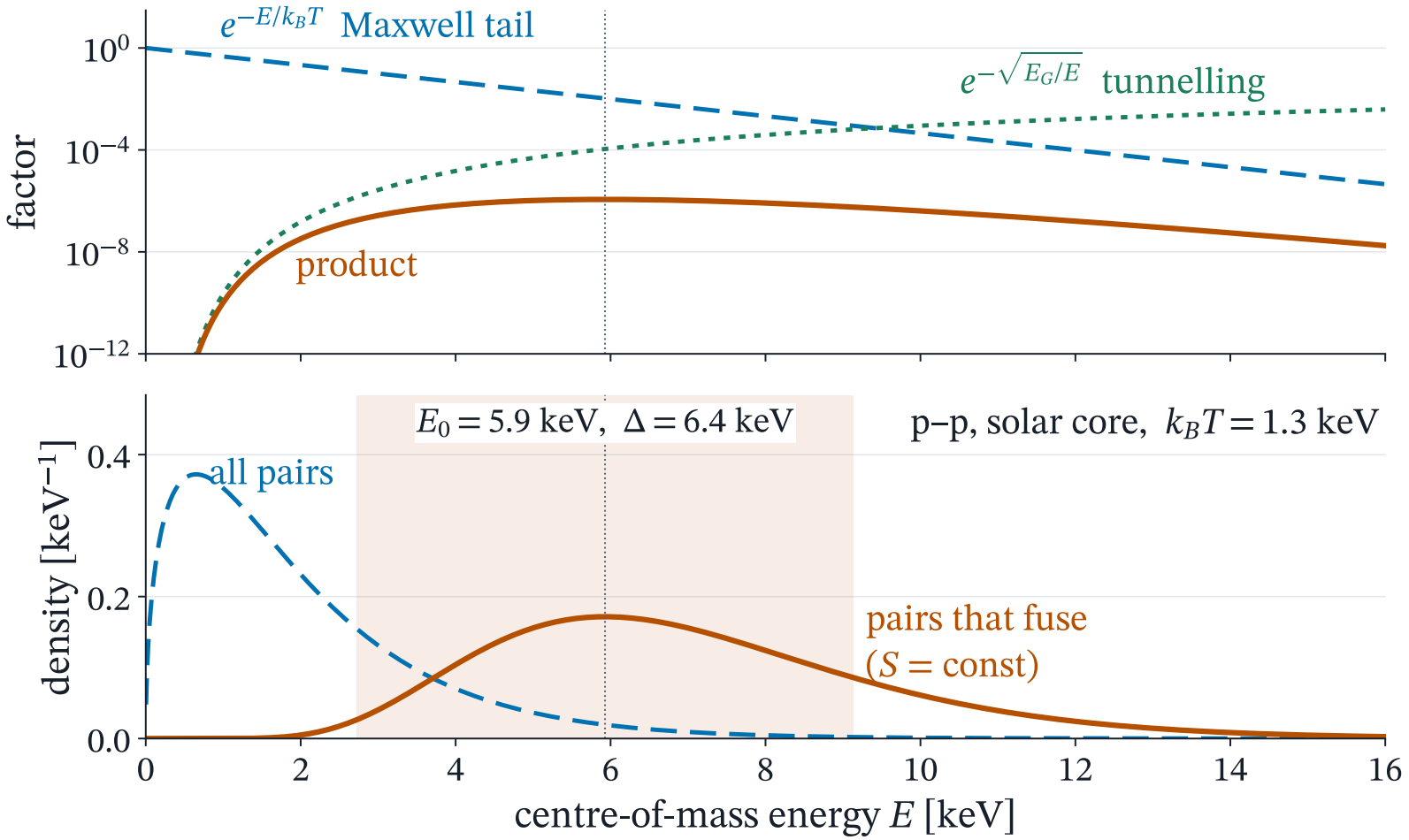
3.3 Three branches



from neutrino fluxes (B16): $\Phi(^7\mathrm{Be})/\Phi(\mathrm{pp}) = f/(2-f), \quad f = \mathrm{pp\ II} + \mathrm{pp\ III}$

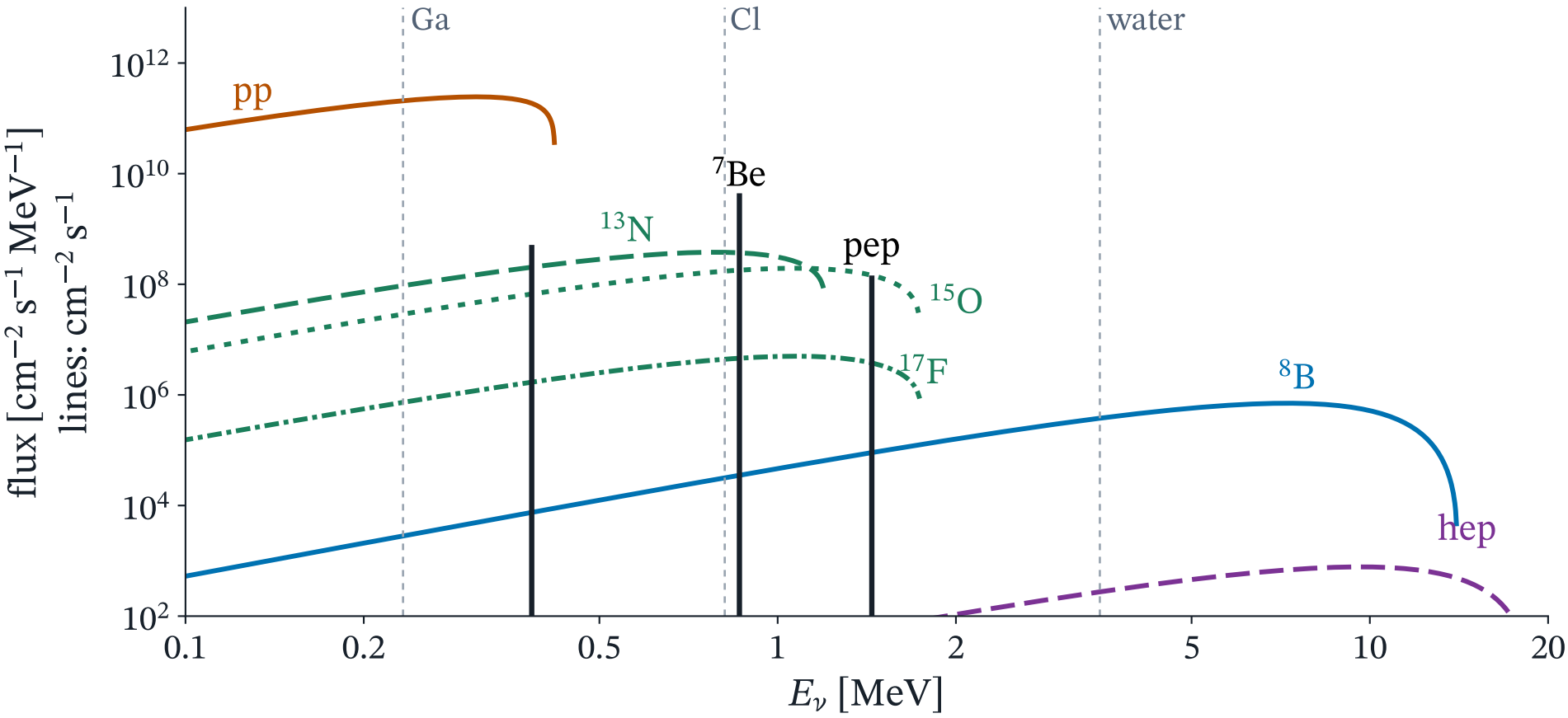
Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011) · Vinyoles et al. · ApJ 835, 202 (2017)

3.3 Solar core



p–p: $E_G = 0.49$ MeV, $k_BT \approx 1.3$ keV $\Rightarrow$ $E_0 \approx 6$ keV $\approx 4.6\, k_BT$

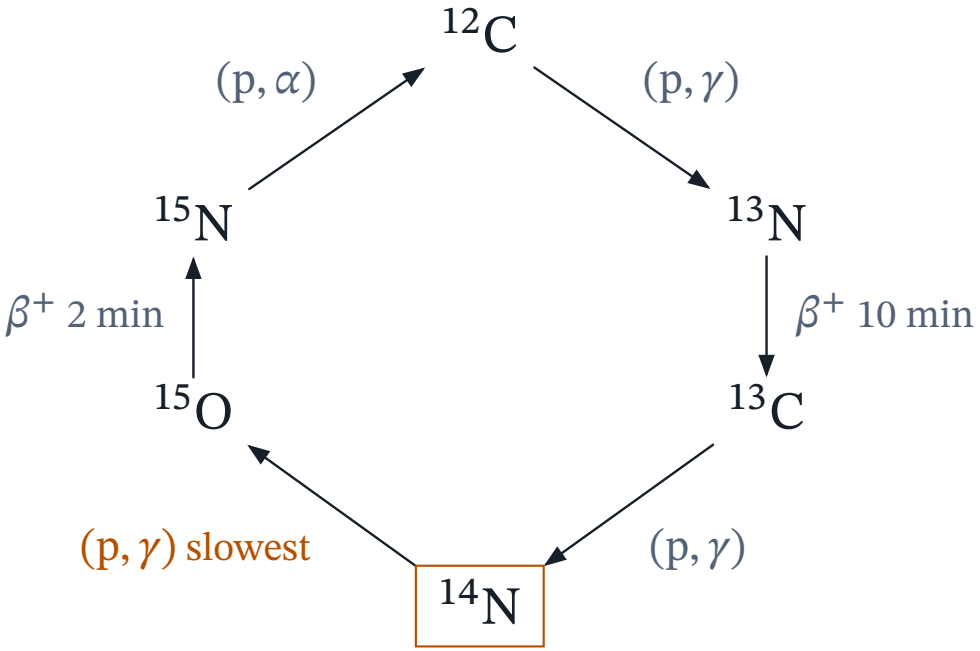
3.3 Solar neutrinos



measured: Cl 1968 · Ga 1990s · water 1996– · pp (Borexino 2014) · CNO (Borexino 2020)

Davis et al. · PRL 20, 1205 (1968) · Borexino · Nature 512, 383 (2014) · Borexino · Nature 587, 577 (2020) · fluxes B16-GS98

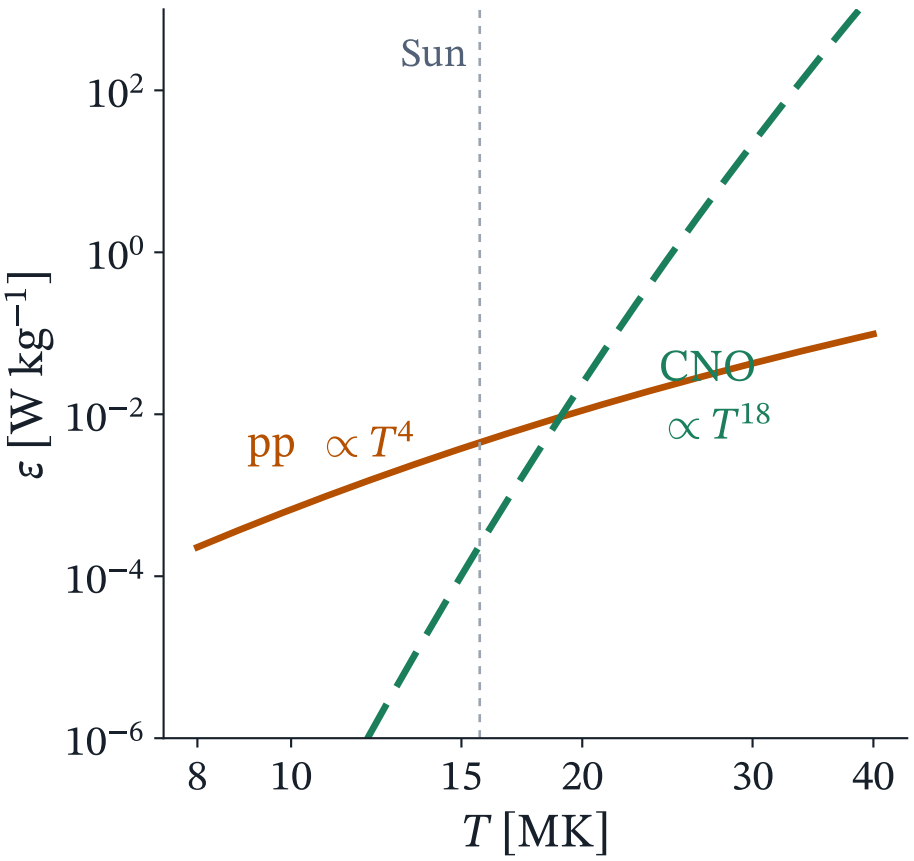
3.4 CNO cycle

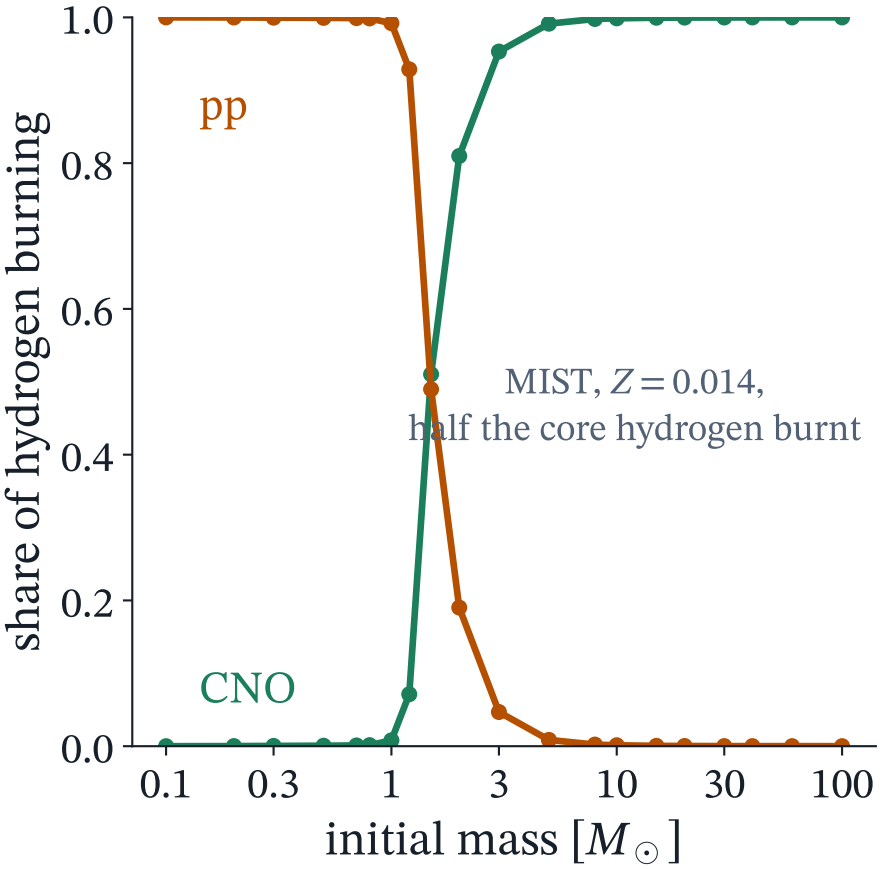


net	$4\,\mathrm{p} \rightarrow {}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e$
energy	26.7 MeV, 25.0 MeV stays
catalyst	C, N, O are not used up
bottleneck	${}^{14}\mathrm{N} + \mathrm{p} : Z_1 Z_2 = 7$

Bethe · Phys. Rev. 55, 434 (1939) · Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011)

3.4 pp or CNO?

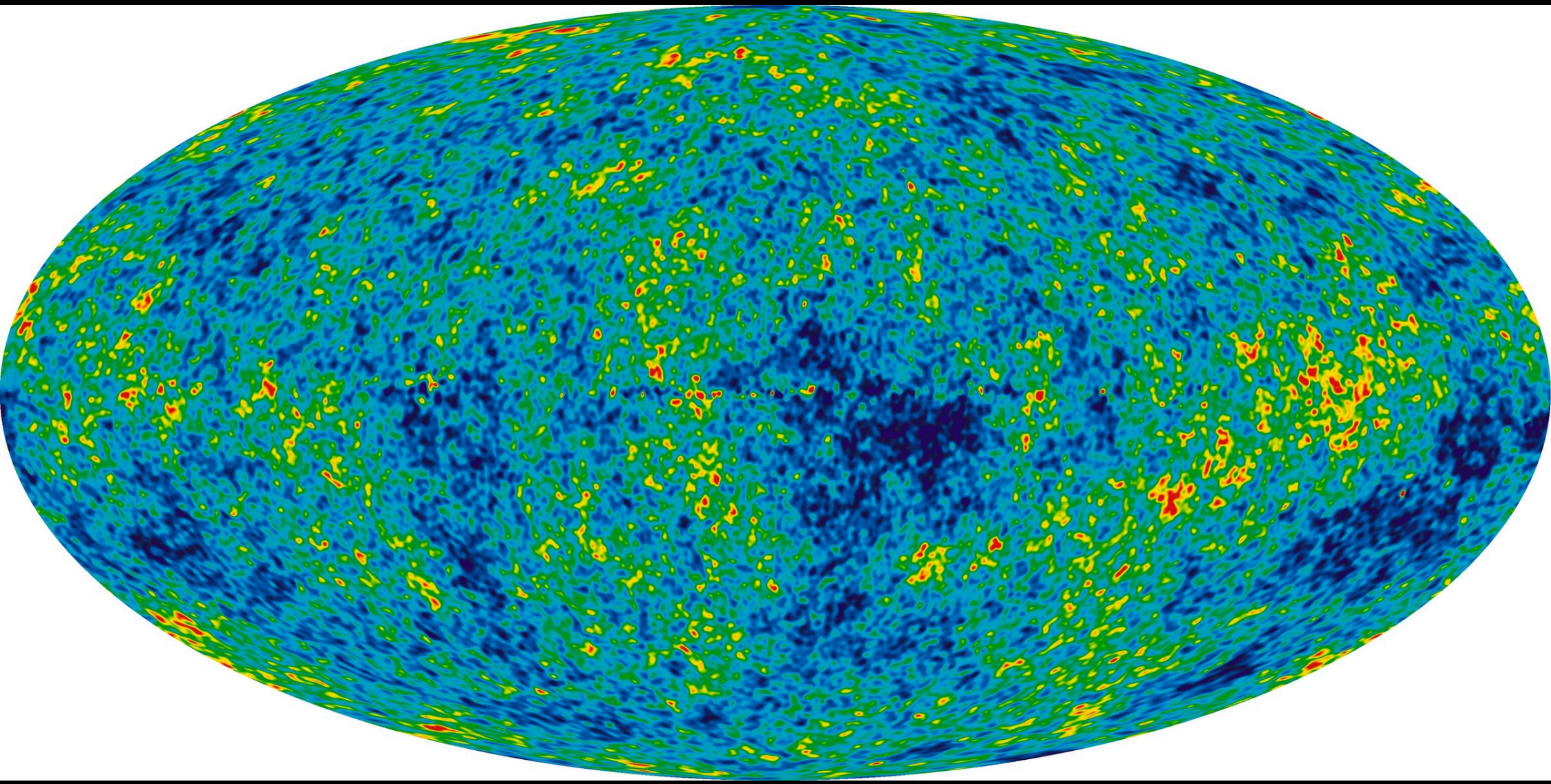




$$\frac{\mathrm{d}\ln\langle\sigma v\rangle}{\mathrm{d}\ln T} = \frac{\tau - 2}{3}, \quad \tau = \frac{3E_0}{k_B T}$$

equal at 19 MK · CNO wins above 1.5 $M_\odot$

$S_{11} = 4.01 \times 10^{-22}$ keV b, $S_{1,14} = 1.66$ keV b (Adelberger et al. 2011) · MIST · Dotter, ApJS 222, 8 (2016)



NASA / WMAP Science Team, public domain

3.5 Neutrons decide the helium

freeze-out $$n/p = e^{-Q/k_B T_f}, \quad Q = 1.293\ \text{MeV}, \quad k_B T_f \approx 0.8\ \text{MeV} \quad \Rightarrow \quad 0.20$$

bottleneck $$n_{\mathrm{D}}/n_{\mathrm{n}} \approx \frac{3}{4}\eta \frac{2\zeta(3)}{\pi^2}\left(\frac{4\pi k_B T}{mc^2}\right)^{3/2} e^{B_{\mathrm{D}}/k_B T} = 1 \quad \Rightarrow \quad k_B T_{\mathrm{D}} = 0.066\ \text{MeV}$$

decay $$n/p = 0.20\, e^{-\Delta t/\tau_n} = 0.14 \quad (\tau_n = 878\ \text{s}, t_{\mathrm{D}} \approx 300\ \text{s})$$

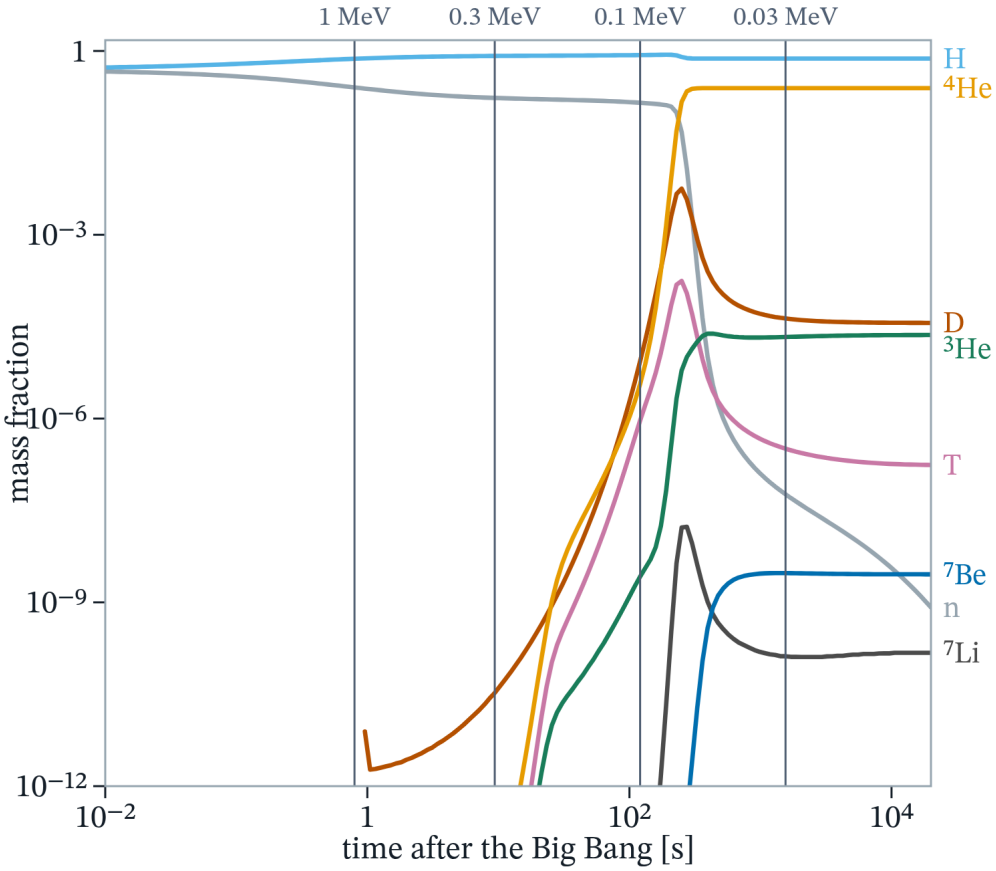
helium $$Y_p = \frac{2n/p}{1 + n/p} = 0.25$$

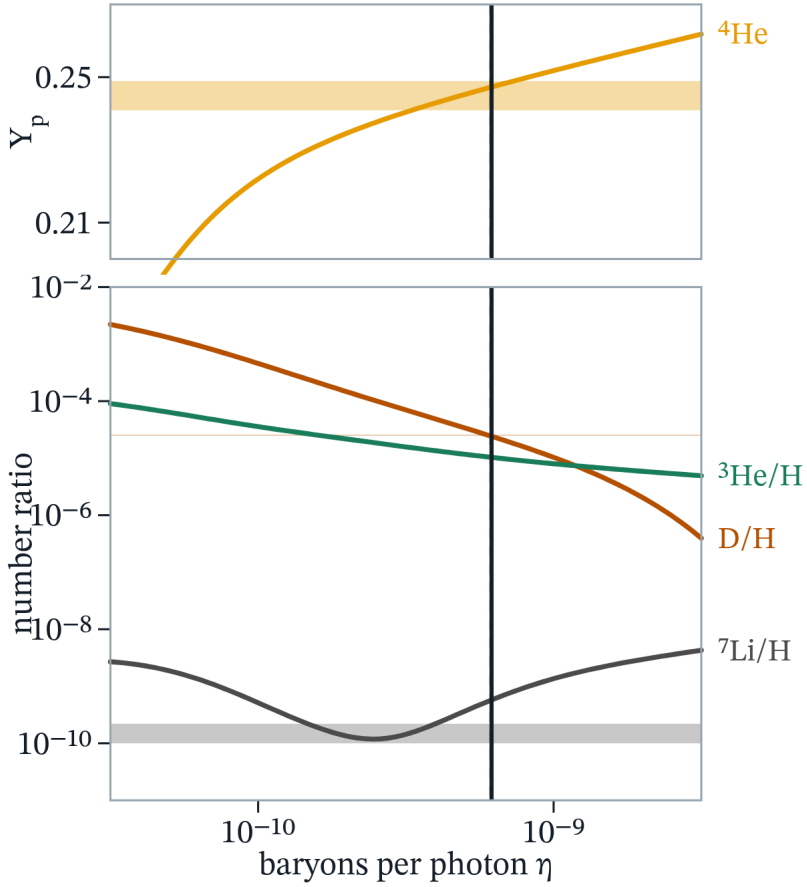
$$Y_p = 0.247$$

network (LINX) · observed 0.245 ± 0.004

Alpher, Bethe & Gamow · Phys. Rev. 73, 803 (1948) · LINX · Phys. Rev. D 112, 063531 (2025) · Aver et al. · JCAP 07, 011 (2015)

3.5 Primordial nucleosynthesis





$Y_p = 0.2471 \cdot \mathrm{D/H} = 2.48 \times 10^{-5} \cdot {}^7\mathrm{Li/H} = 5.57 \times 10^{-10}$

bands: observed ($\pm 2\sigma$ for D, Li), dashed: η from the CMB (Planck) · LINX, PRIMAT 2023 network

η $\eta = 6.17 \times 10^{-10}$

3.5 Why it stops at lithium

no stable $A = 5, 8$	$^5\mathrm{He}, ^5\mathrm{Li} : 10^{-21}$ s, $^8\mathrm{Be} : 8 \times 10^{-17}$ s
dilute	$\rho_b \approx 10^{-2}$ kg/m^3 at $k_B T = 0.07$ MeV (Sun: 1.5×10^5)
fast	expansion: $T \propto t^{-1/2}$, over after about 20 min
result	H 75 %, ^{4}He 25 %, D/H 2.5×10^{-5}, ^{7}Li/H 5×10^{-10}

observed $^7\mathrm{Li/H} = 1.6 \times 10^{-10}$

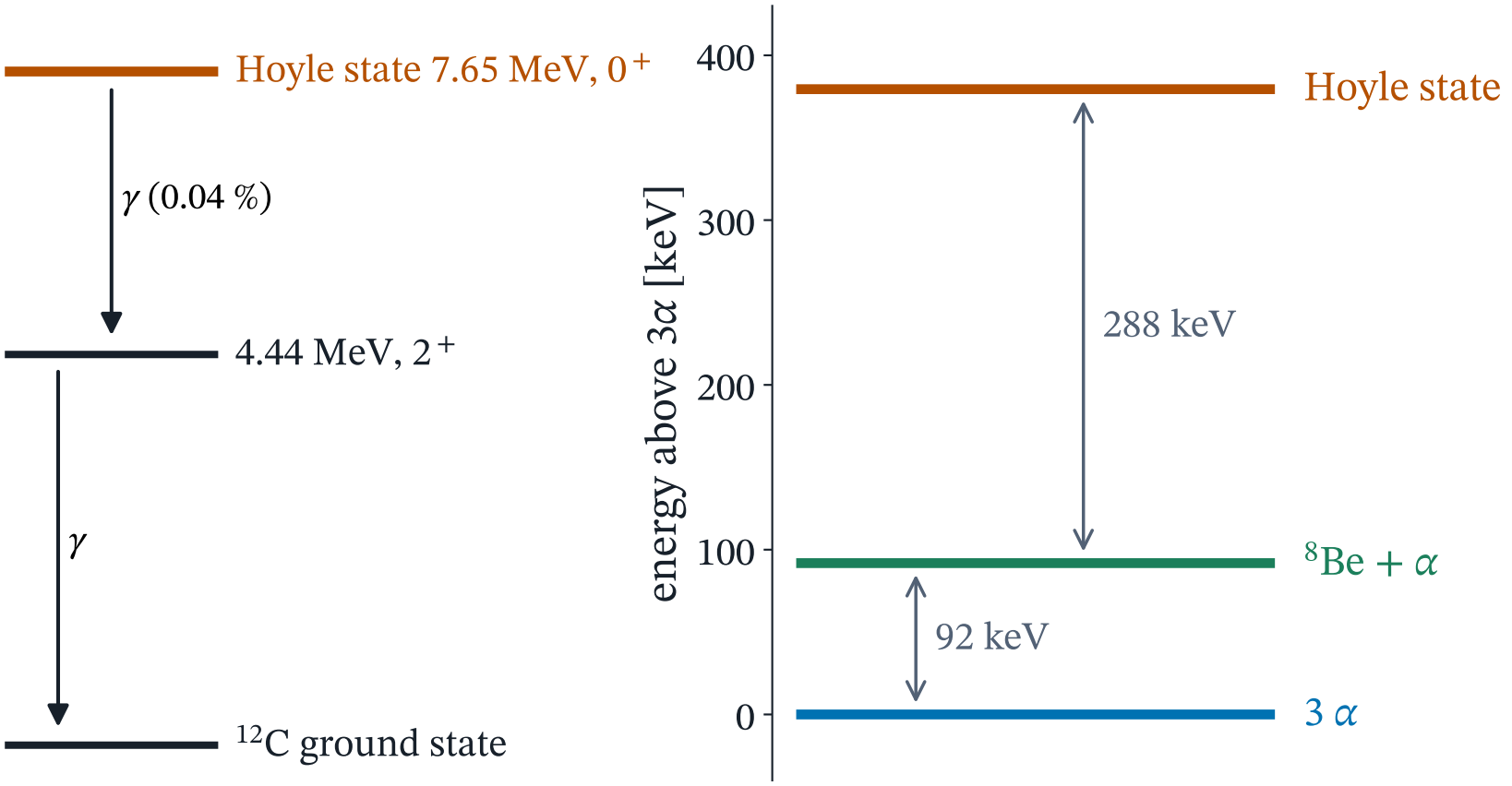
the lithium problem

Cooke et al. · ApJ 855, 102 (2018) · PDG BBN review (2024)



NASA, ESA, N. Smith (UC Berkeley), Hubble Heritage Team (STScI/AURA), CC BY 4.0

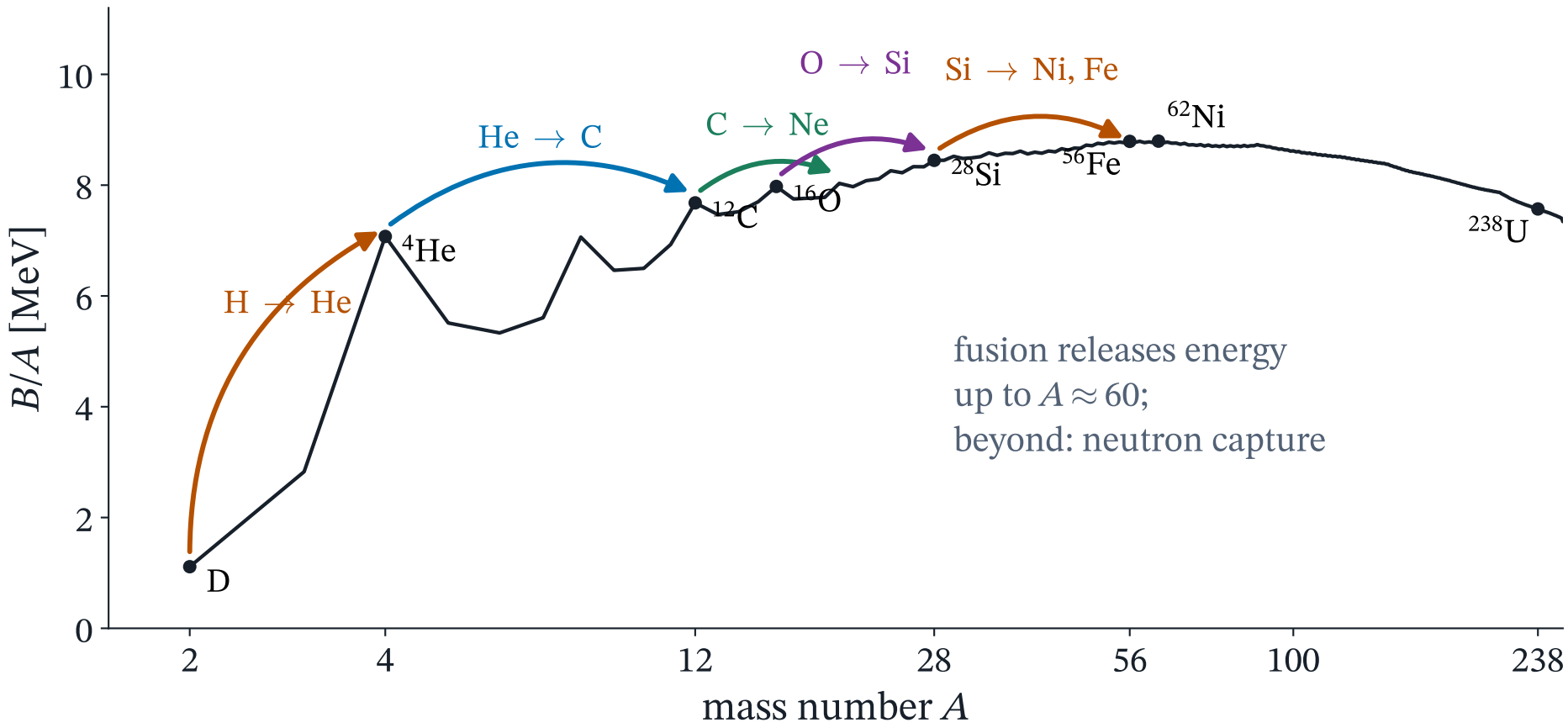
3.6 Triple alpha and the Hoyle state



$$^4\mathrm{He} + {}^4\mathrm{He} \rightleftarrows {}^8\mathrm{Be}, \quad {}^8\mathrm{Be} + {}^4\mathrm{He} \rightarrow {}^{12}\mathrm{C}^* \rightarrow {}^{12}\mathrm{C} + \gamma, \quad {}^{12}\mathrm{C} + {}^4\mathrm{He} \rightarrow {}^{16}\mathrm{O} + \gamma$$

Hoyle · ApJS 1, 121 (1954) · Cook, Fowler, Lauritsen & Lauritsen · Phys. Rev. 107, 508 (1957) · AME2020

3.6 Binding energy per nucleon

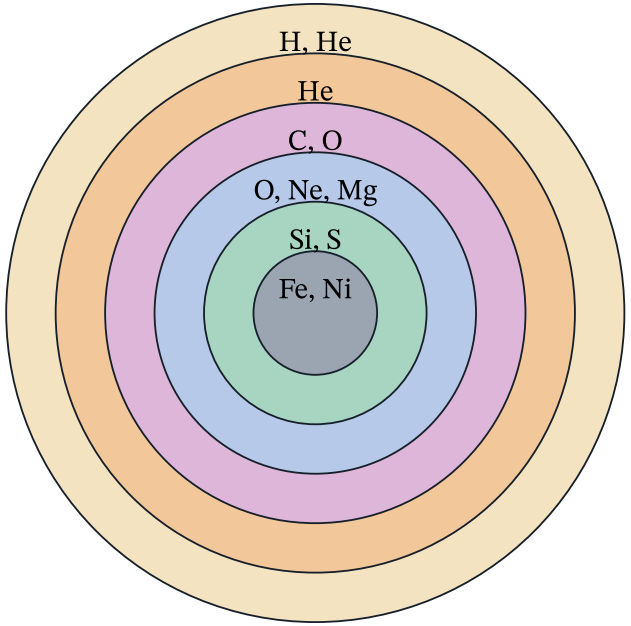


most bound: ^{62}Ni 8.795 MeV ^{56}Fe 8.790 MeV ^{4}He 7.074 MeV

AME2020 · Wang et al. · Chin. Phys. C 45, 030003 (2021)

3.6 Burning stages of a massive star

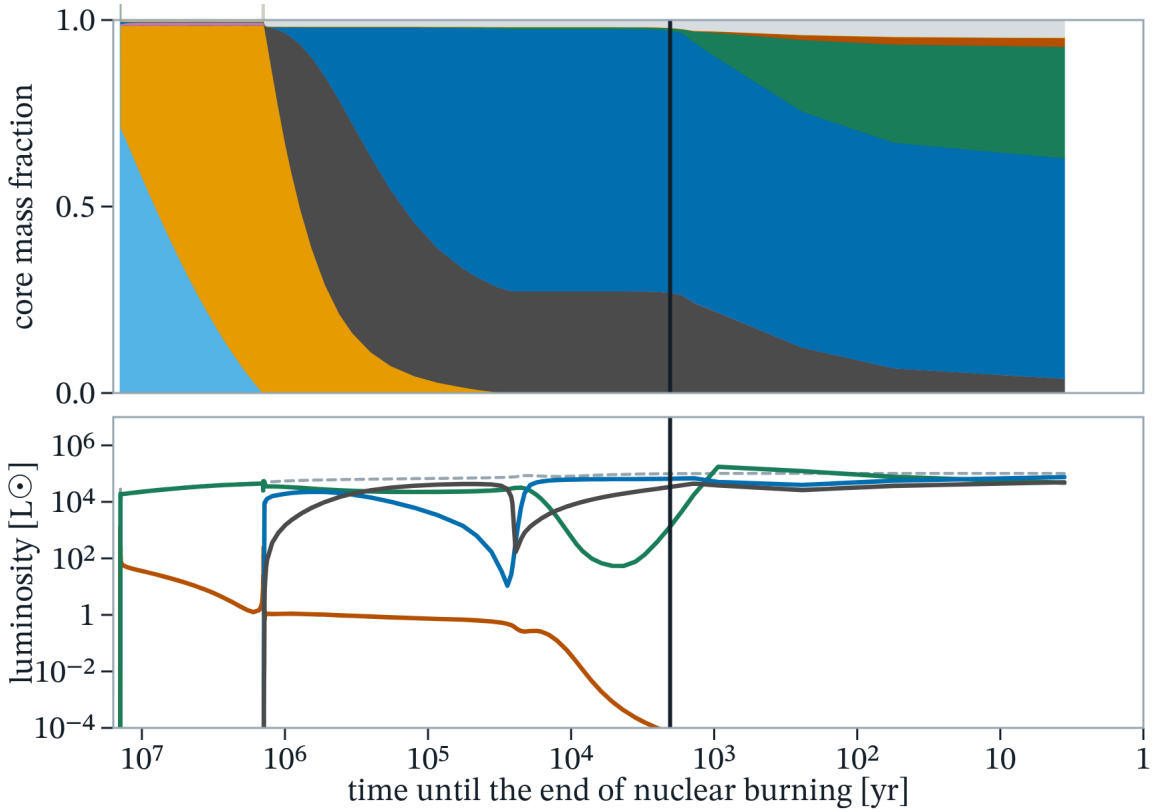
fuel	ash	T [GK]	$k_B T$ [keV]	duration
H	He	0.037	3	8.1 Myr
He	C, O	0.19	16	1.2 Myr
C	Ne, Mg	0.87	75	980 yr
Ne	O, Mg	1.6	140	0.6 yr
O	Si, S	2.0	170	1.3 yr
Si	Fe, Ni	3.3	280	11 d

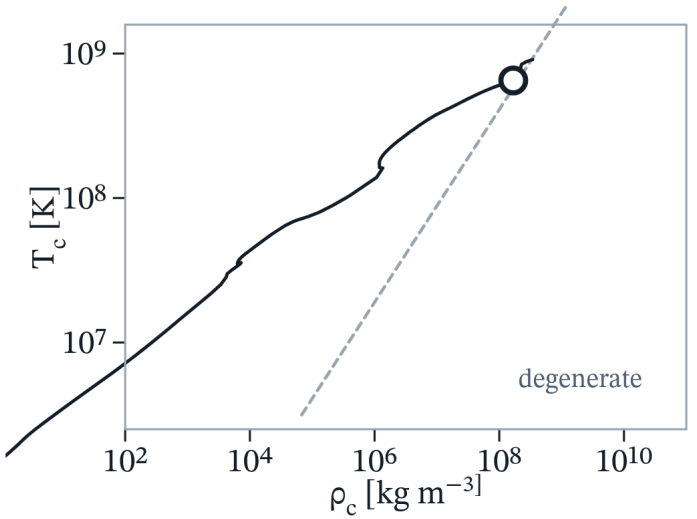


20 $M_\odot$ after He: neutrino losses $\propto T^9$

Heger, Woosley, Fryer & Langer · Massive star evolution through the ages (2003), Table 1 · shells not to scale

3.6 Build a star





shell burning around an inert core · M = 11.56 M⊙

T_c = 652 MK = 56.15 keV, $\rho_c = 1.7 \times 10^8$ kg m^{-3}

end: C exhausted; Ne, O, Si burn within about 2 years, the Fe core collapses: supernova, neutron star or black hole

then (20 M⊙ model): Ne 0.6 yr, O 1.3 yr, Si 11 d

H He C N O Ne Mg Si

pp chain CNO cycle triple alpha carbon burning

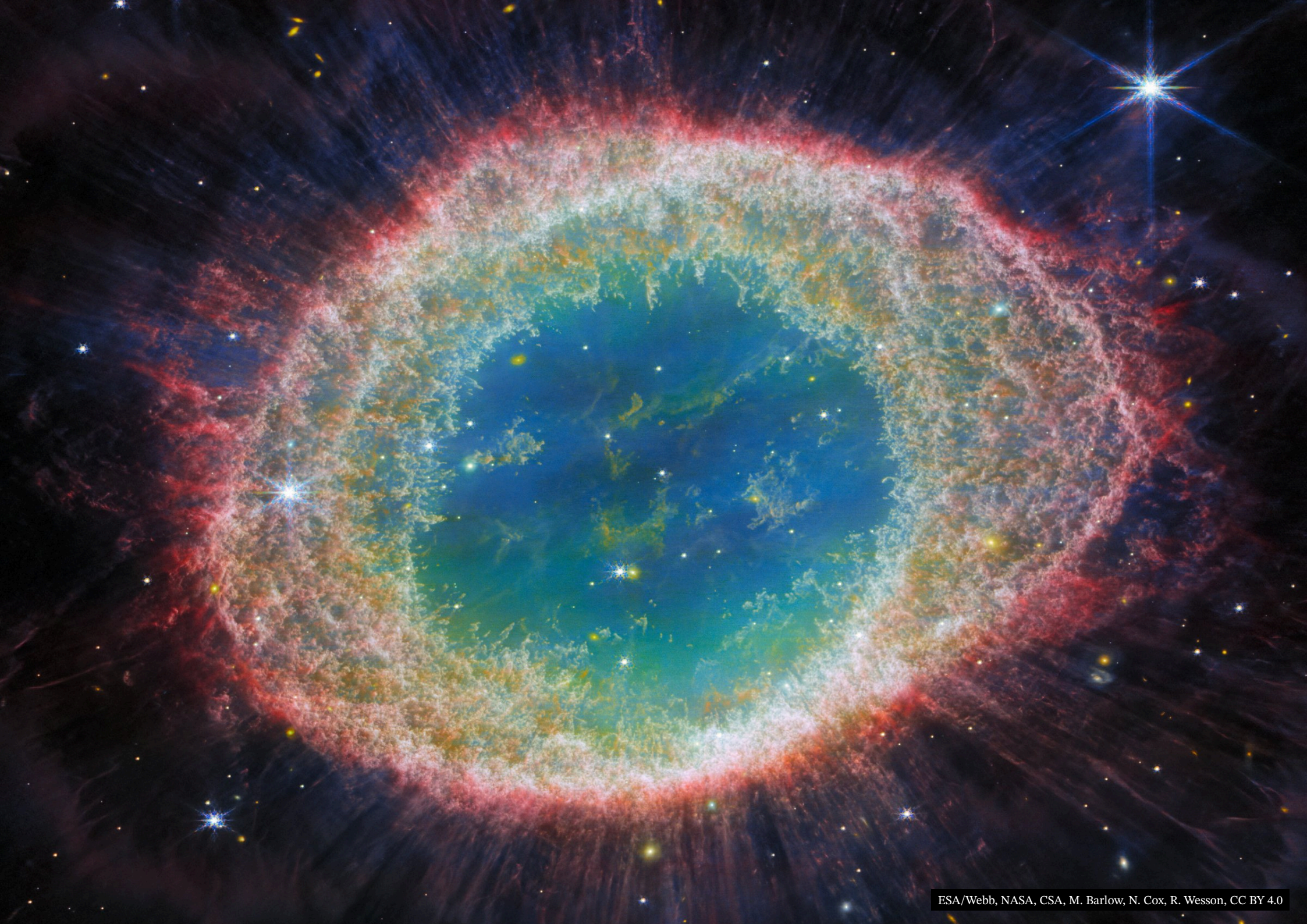
MIST v1.2 (MESA) tracks · Dotter 2016, Choi et al. 2016

initial mass — 15 M⊙

metals — ○ Z = 0.00014 ○ Z = 0.001 ◉ Z = 0.014 (Sun) ○ Z = 0.045

age — 14.17 Myr

run — ☐



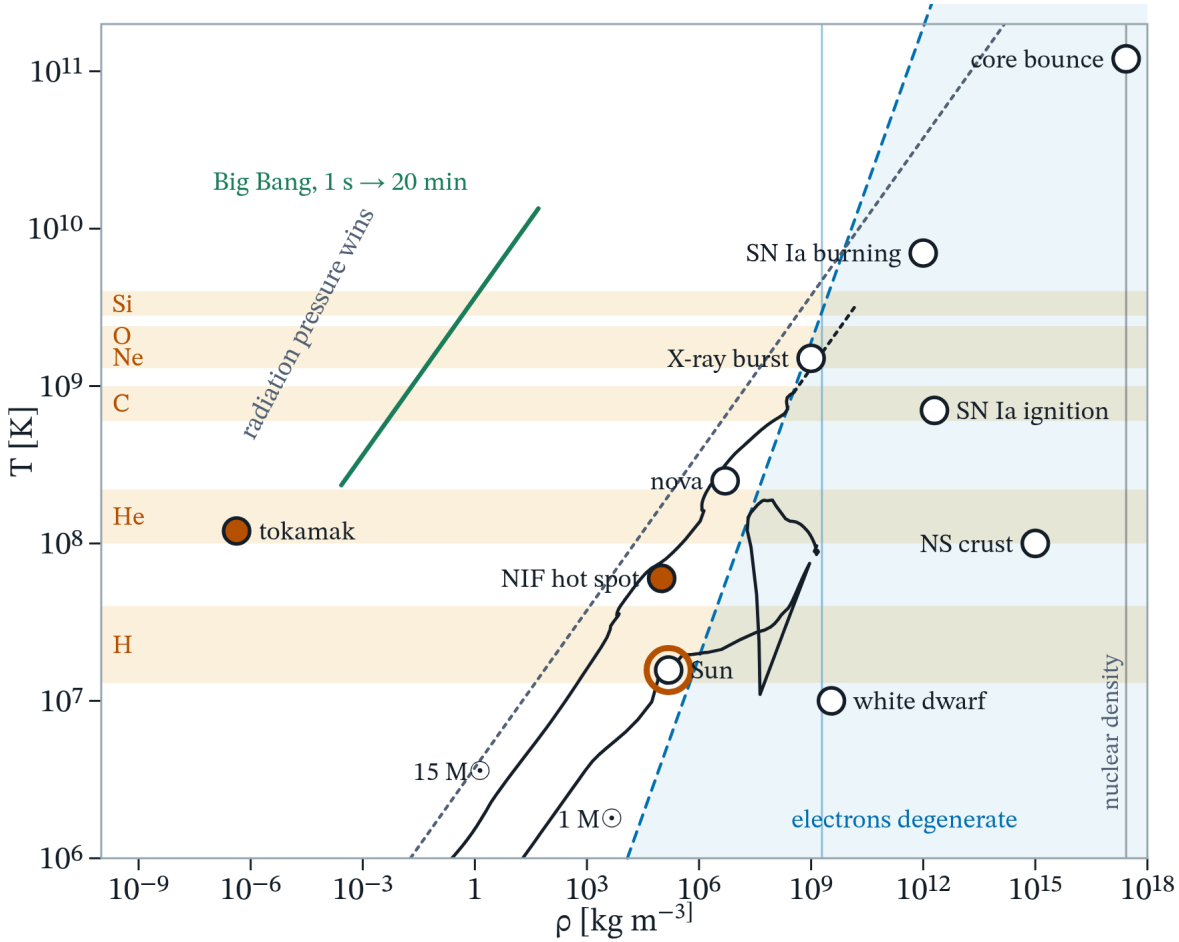
ESA/Webb, NASA, CSA, M. Barlow, N. Cox, R. Wesson, CC BY 4.0

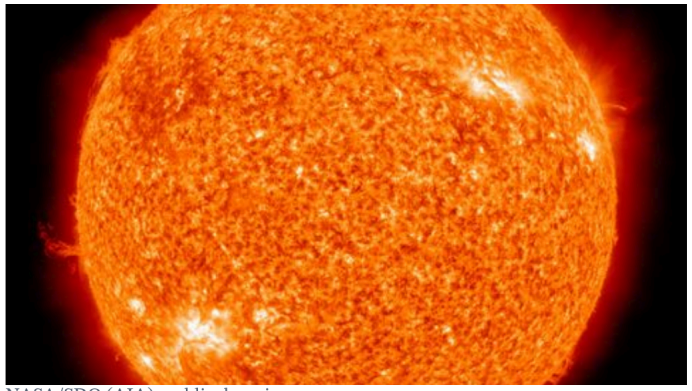
3.7 Degeneracy ends the burning

degenerate	$E_F = k_B T$: pressure no longer depends on T, no thermostat
lightest star	$0.075 M_\odot$: degeneracy stops the contraction before p–p ignites
brown dwarf	$0.013 M_\odot \approx 13 M_{\mathrm{Jupiter}}$: deuterium burning only
white dwarf	$0.6 M_\odot$: $\rho_c = 3.5 \times 10^9$ kg/m^3, $R \approx 8700$ km
Chandrasekhar	$M_{\mathrm{Ch}} = \dfrac{\sqrt{3\pi}}{2}\,\omega_3 \left(\dfrac{\hbar c}{G}\right)^{3/2} \dfrac{1}{(\mu_e m_u)^2} = 1.46 M_\odot \quad (\mu_e = 2)$
neutron star	$1.4 M_\odot$, $R \approx 12$ km, nuclear density

Chandrasekhar · ApJ 74, 81 (1931) · Chabrier & Baraffe · ARA&A 38, 337 (2000) · Spiegel et al. · ApJ 727, 57 (2011)

3.8 Where fusion happens





NASA/SDO (AIA), public domain

Sun, centre

$\rho \approx 1.5 \times 10^{5}$ kg m^{-3}, $T \approx 1.6 \times 10^{7}$ K

pp chain (99 %), CNO (1 %)

Model S, Christensen-Dalsgaard et al., Science 272, 1286 (1996), doi:10.1126/science.272.5266.1286

orange bands: burning temperatures (20 M☉, Heger et al. 2003) · solid: MIST centres · dotted: $\rho \propto T^3$

◉ Sun ○ white dwarf ○ nova ○ X-ray burst ○ SN Ia ignition ○ SN Ia burning ○ core bounce ○ NS crust ○ tokamak ○ NIF hot spot

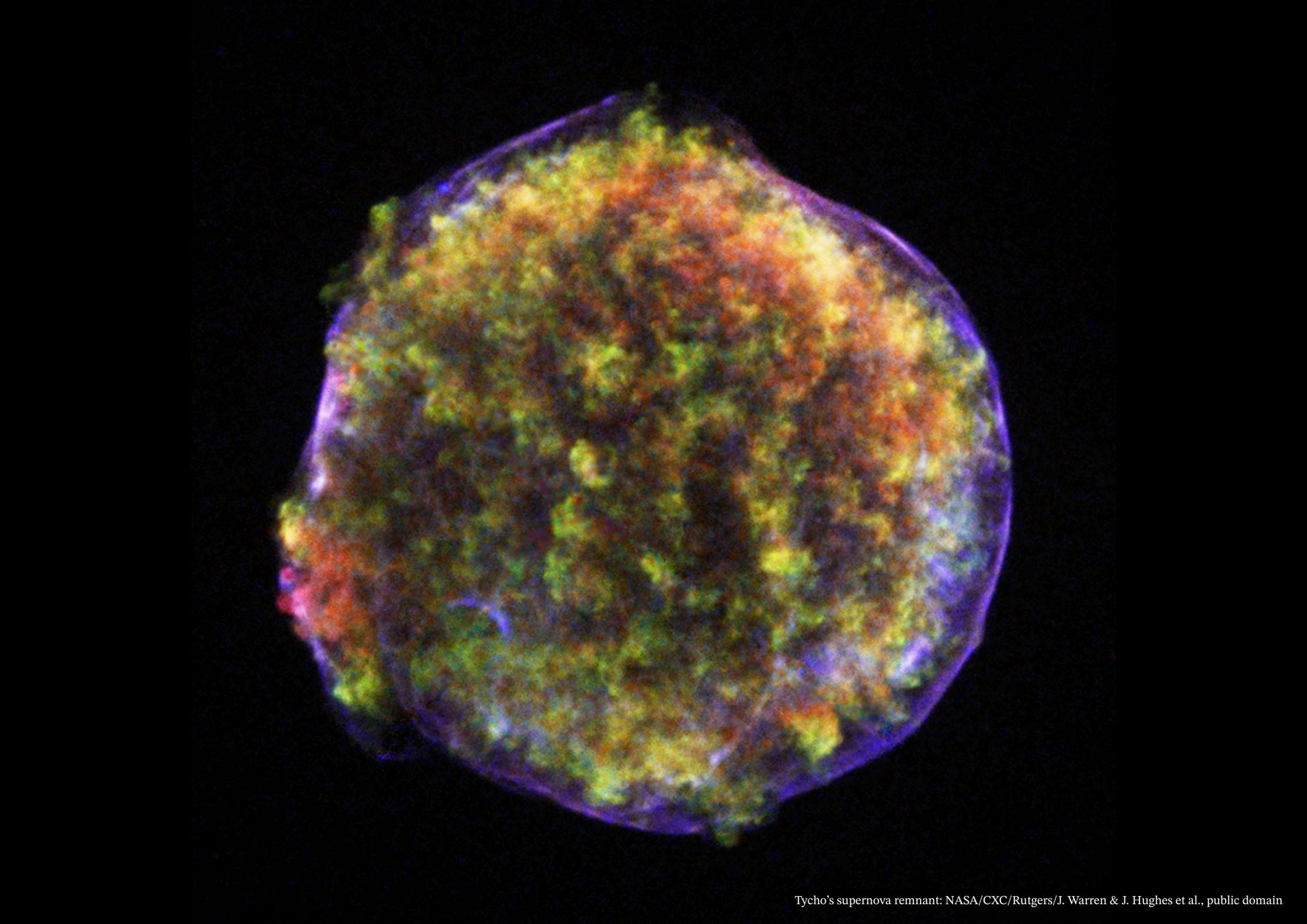


GK Persei: NASA/CXC/RIKEN/D. Takei et al., public domain

3.8 Novae: hot CNO on a white dwarf

system	white dwarf accretes hydrogen from a companion
ignition	degenerate layer: heating without expansion, runaway
hot CNO	$T \approx 2 - 3 \times 10^8$ K: capture faster than β^+ decay of ^{14}O (71 s), ^{15}O (122 s)
ejecta	$10^{-5} - 10^{-4} M_\odot$, rich in ^{13}C, ^{15}N, ^{17}O

José & Hernanz · ApJ 494, 680 (1998)



Tycho’s supernova remnant: NASA/CXC/Rutgers/J. Warren & J. Hughes et al., public domain

3.8 Type Ia supernova: a carbon bomb

progenitor C/O white dwarf grows towards M_{Ch}

ignition $\rho_c \approx 2 \times 10^{12}$ kg/m^3, degenerate: no thermostat

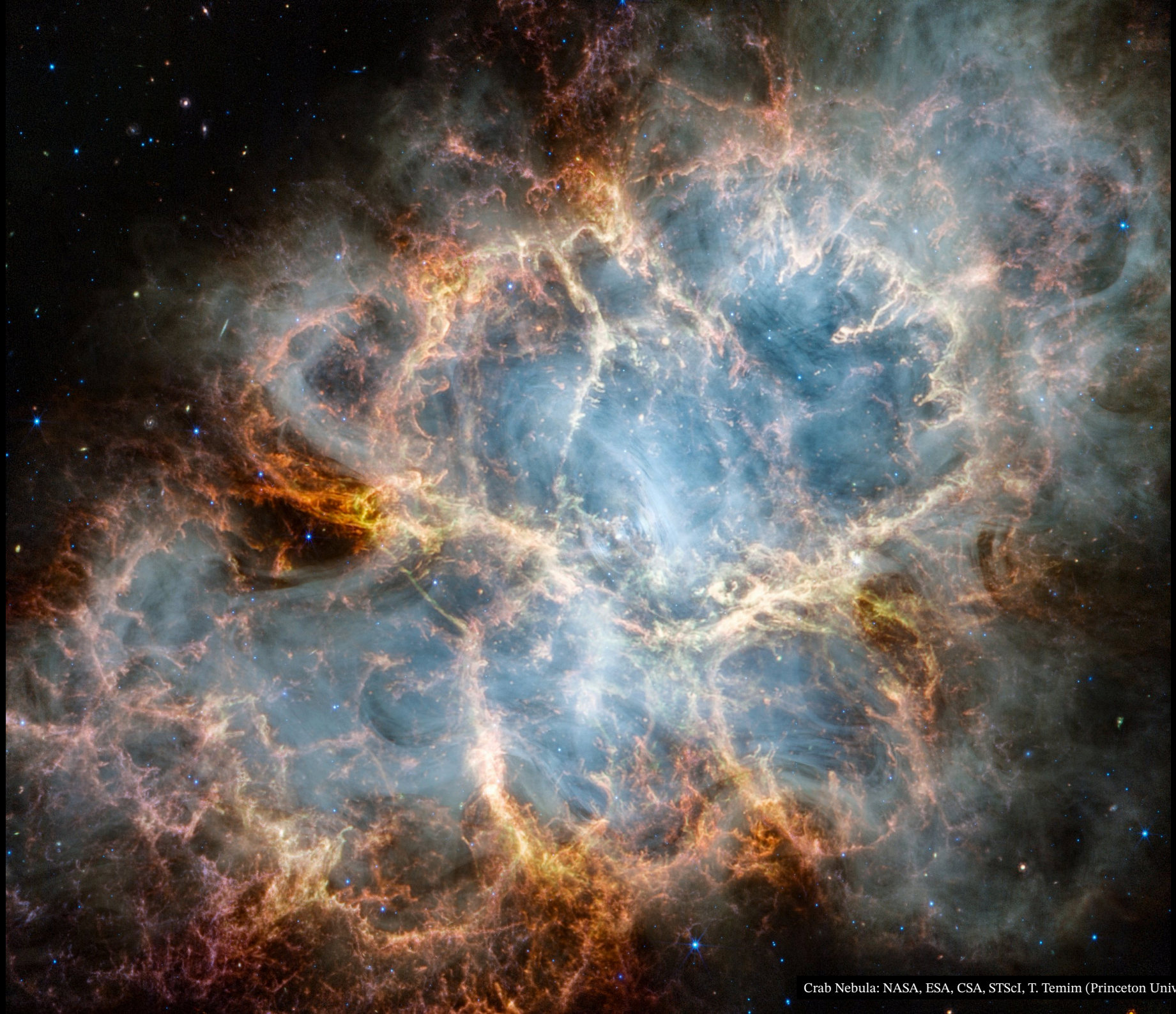
energy C, O $\rightarrow$ ^{56}Ni: 0.81 MeV per nucleon $\times\, 1.4 M_\odot = 2 \times 10^{44}$ J $>$ binding

light $^{56}\mathrm{Ni} \xrightarrow{6.1\,\mathrm{d}} {}^{56}\mathrm{Co} \xrightarrow{77\,\mathrm{d}} {}^{56}\mathrm{Fe}$

fusion unbinds the whole star

no remnant · about 0.6 $M_\odot$ of ^{56}Ni

Hillebrandt & Niemeyer · ARA&A 38, 191 (2000)



Crab Nebula: NASA, ESA, CSA, STScI, T. Temim (Princeton University), CC BY 4.0

3.8 Core collapse: gravity, not fusion

iron core	no fusion energy left: B/A is maximal
collapse	photodisintegration, electron capture; bounce at nuclear density
energy	$\frac{3}{5}\frac{GM^2}{R} = 3 \times 10^{46}$ J (1.4 $M_\odot$, 12 km), 99 % neutrinos
SN 1987A	24 neutrinos in Kamiokande-II, IMB, Baksan; $0.07 M_\odot$ of ^{56}Ni

fusion builds the core and burns the ejecta

explosive Si and O burning

Hirata et al. · PRL 58, 1490 (1987) · Bionta et al. · PRL 58, 1494 (1987) · Arnett et al. · ARA&A 27, 629 (1989) · Janka et al. · Phys. Rep. 442, 38 (2007)

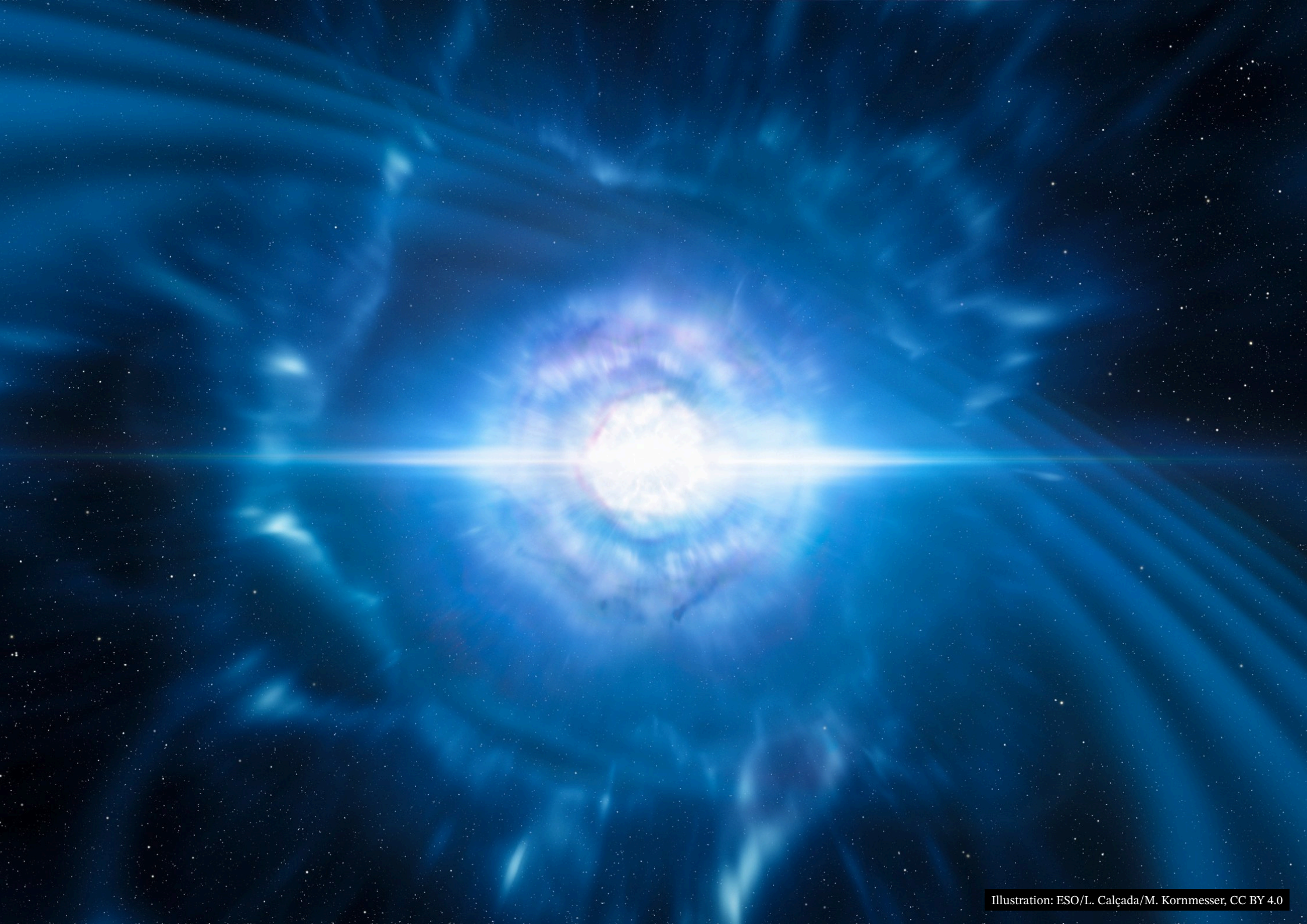


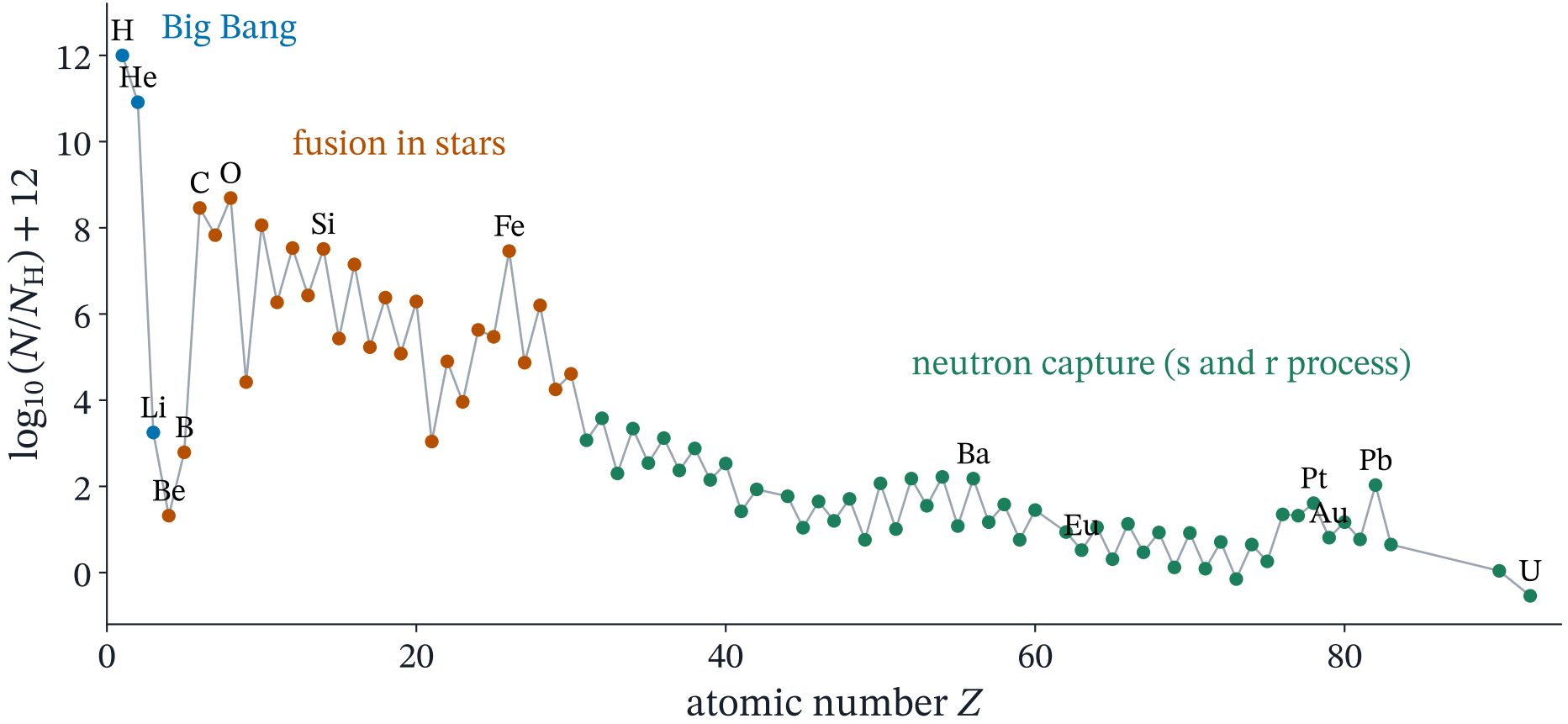
Illustration: ESO/L. Calçada/M. Kornmesser, CC BY 4.0

3.8 Neutron stars: bursts, crusts, mergers

X-ray burst	accreted H, He ignite: He flash, rapid proton capture up to Sn–Te in seconds
crust	pycnonuclear fusion: zero-point motion in a lattice, density instead of temperature
merger	GW170817: neutron capture (r process), kilonova · not fusion
on Earth	muon-catalysed fusion is fusion by density too (Chapter 9)

Schatz et al. · PRL 86, 3471 (2001) · Haensel & Zdunik · A&A 227, 431 (1990) · LIGO/Virgo · PRL 119, 161101 (2017)

3.8 Where the elements come from



Asplund, Amarsi & Grevesse · A&A 653, A141 (2021), CC BY 4.0 · Burbidge, Burbidge, Fowler & Hoyle · Rev. Mod. Phys. 29, 547 (1957)

3.8 Stars and fusion on Earth

	star	reactor
confinement	gravity, $\tau_E \sim 10^7$ yr	magnetic field, $\tau_E \sim$ s
fuel	p–p, $S \sim 10^{-22}$ keV b	D–T, $S \sim 10^4$ keV b
temperature	1.35 keV	10–20 keV
power density	262 W/m^3	$\sim$ MW/m^3
control	negative heat capacity	heating and feedback

Credits

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sdo.gsfc.nasa.gov

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17 NASA, ESA, N. Smith, Hubble Heritage Team, CC BY 4.0
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chandra.harvard.edu/photo/2015/gkper

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chandra.harvard.edu

29 NASA, ESA, CSA, STScI, T. Temim, CC BY 4.0
esawebb.org/images/weic2326a

31 ESO/L. Calçada/M. Kornmesser, illustration, CC BY 4.0
eso.org

Data: MIST (Dotter 2016; Choi et al. 2016), LINX/PRIMAT, Model S, AME2020, Asplund et al. 2021, B16, Heger et al. 2003. Plots, widgets, calculations: Christopher Albert, CC BY 4.0.

AI tools assisted drafting and layout; scientific review by Christopher Albert.